A Comparative Machine Learning Approach for Predicting Insomnia and Sleep Quality Degradation among Bangladeshi University Students

Jaky Ahmed Chowdhury
Dept of Computer Science and Engineering
BRAC University
Dhaka, Bangladesh
https://orcid.org/0009-0004-8615-9406

Md. Khairul Ashik
Dept of Computer Science and Engineering
BRAC University
Dhaka, Bangladesh
md.khairul.ashik@g.bracu.ac.bd

Farhana Tasnim Eva
Dept of Computer Science and Engineering
BRAC University
Dhaka, Bangladesh
farhana.tasnim.eva@g.bracu.ac.bd

***Abstract*—Sleep disorders such as insomnia and sleep apnea, are really common among university students. This is often because of their irregular schedules, academic stress, and excessive use of screens late at night. However, most of these cases go undiagnosed because the usual methods of diagnosis, such as polysomnography or specialist consultations, are expensive and time-consuming. This is especially true in countries like Bangladesh where resources are limited. There's a need for a low-cost and scalable way to screen for sleep disorders that can identify individuals who are at risk using data that's easy to collect, such as lifestyle and demographic information. But, there's a lack of large datasets on sleep disorders among university students, which makes it hard to create reliable data-driven screening models. To address this issue, a survey was conducted among university students in Bangladesh, and the responses were combined with an existing dataset to create a larger dataset of 811 records. This dataset included information about demographics, behavior, lifestyle, and clinical attributes. Seven different classifiers, including Decision Tree, Random Forest, AdaBoost, Gradient Boosting, Logistic Regression, XGBoost, and CatBoost, were trained and compared using stratified train/test splitting, SMOTE-based class balancing, and grid search with 5-fold stratified cross-validation. The results showed that CatBoost performed the best, with a test accuracy of 71.6% and a macro F1 score of 0.72, followed closely by Gradient Boosting with an accuracy of 69.7%. Sleep duration and stress level emerged as the strongest predictors of sleep disorders. Sleep disorders can have serious consequences on a student's academic performance, mental health, and overall well-being. Early identification and intervention can help mitigate these effects and improve the quality of life for university students.**

***Index Terms*—Sleep disorders, insomnia, sleep apnea, machine learning, university students, comparative analysis**

I. Introduction

Sleeping well is crucial for our overall health, but many people struggle with it. When we don't sleep properly, it can lead to problems like insomnia and sleep apnea. Normally, doctors use special equipment to diagnose these issues, but it can be costly and hard to use for a large number of people, like university students. Machine learning offers a new way to identify sleep disorders by looking at patterns in people's lives, such as their demographics, behavior, lifestyle, and clinical information. This approach can help find connections between these factors and sleep disorders. As a result, machine learning models can be a useful tool for initial screenings, helping to pinpoint students who might be at a higher risk of sleep disorders. By using this method, we can potentially make it easier to detect sleep problems early on, which is important for getting proper treatment and improving overall well-being. It's not a replacement for a doctor's diagnosis, but rather a way to simplify the process and make it more accessible to more people. The major contributions of this project are:

- Expansion of an existing Bangladeshi university sleep-disorder dataset through additional data collection.
- Comprehensive exploratory data analysis of demographic, behavioral, lifestyle, and clinical characteristics.
- Development of an updated preprocessing pipeline for mixed categorical and numerical survey data.
- Numerical transformation of range-based Sleep Duration and Daily Steps variables.
- Conversion of Blood Pressure into separate systolic and diastolic numerical features.
- Class balancing of the training data with SMOTE and hyperparameter tuning of tree-based models with grid search under stratified cross-validation.
- Comparison of seven machine learning algorithms for three-class sleep-disorder classification, together with confusion-matrix and feature-importance analysis of the results.

II. Related Work

Alazaidah et al. [1] investigated machine learning approaches for sleep-disorder prediction using both regression and classification methods. The researchers evaluated a large number of models using existing sleep-related datasets and demonstrated the potential of machine learning for identifying sleep-related patterns. However, the study primarily depended

on existing datasets and did not perform extensive external validation.

Ravikumar and Bhansali [2] had an idea to use Random Forest. They were fairly correct in that; they worked out, about 96.7% of the time. But, they only looked at 374 people, and only considered one type of classifiers, which may not be There is sufficient information available to get the complete picture.

Kumar et al. [3] reported that some machine learning tools are available to assist. really good at spotting patterns in data related to sleep. They evaluated eight tools found that the ones that work together as a team are especially effective. However, this This study has been restricted due to the availability of a small sample of respondents who were only from the It was one source, and few tests were performed on external data.

Goodyear and Alsammani [4] looked at how well different machine learning classifiers worked to predict sleep disorders. Random Forest worked really well. They also discovered that age, weight category, blood pressure, and job type were all important factors in predicting sleep disorders. But there was a limitation to their study - they only used a small dataset from one source, which might not be representative of the bigger picture. This means their findings might not be totally accurate or applicable to everyone.

Adeyemi [5] created a system to classify things into many groups using three different methods: Random Forest, XGBoost, and LightGBM. The study used a technique called SMOTEENN to make sure the groups had a similar number of examples. The results showed that LightGBM was very good at correctly classifying things, with high precision and recall. However, the study was limited because it only used a small public dataset and didn't test the system with other data to see if it would work just as well.

Alshammari [6] investigated sleep-disorder classification using k-nearest neighbors, Support Vector Machine, Decision Tree, Random Forest, and Artificial Neural Network models with genetic-algorithm-based hyperparameter optimization. The research demonstrated that model performance can vary considerably depending on the algorithm and optimization strategy. Taspinar and Cinar [7] used a dataset that focuses on sleep health and lifestyle. The methods they tested were Random Forest, SVM, Logistic Regression, and kNN. To see how well these methods worked, they used a technique called 10-fold cross-validation. The results showed that Random Forest was the most accurate, getting it right 91.66% of the time. SVM and Logistic Regression were close behind, with accuracy rates of 90.27%. kNN had a lower accuracy rate of 87.50%.

Taher and Ayon [8] compared Random Forest, AdaBoost, Logistic Regression, and Gradient Boosting for sleep-disorder classification using a dataset containing 375 entries. Gradient Boosting achieved the highest accuracy of 93.80%. Alghwiri et al. [9] studied what affects the sleep quality of university students in Jordan. They looked at 1,600 students and found that a certain method called Random Forest was the most accurate way to predict poor sleep quality, getting it right about 74% of the time. This method was very good at finding students who had poor sleep quality, but not as good at finding those who had good sleep quality. One of the limitations of this study is that it only looked at a snapshot in time, so we can't say for sure what causes poor sleep quality. Also, the students reported their own sleep habits, which might not be entirely accurate.

III. DATASET

A. Existing Dataset

Researchers studied sleep problems among university students in Bangladesh. They collected data from five public universities: Hajee Mohammad Danesh Science & Technology University, University of Dhaka, University of Rajshahi, University of Chittagong, and Khulna University of Engineering & Technology. The study used online forms and in-person interviews to gather information from 619 students [10].

B. Newly Collected Dataset

To make the dataset more comprehensive, we gathered more survey responses. This updated dataset adds 192 more responses, which will help improve the accuracy of the findings.

C. Combined Dataset

The existing data and new information were combined into one set. This combined set was then mixed up using a fixed random code to make sure the results could be repeated.

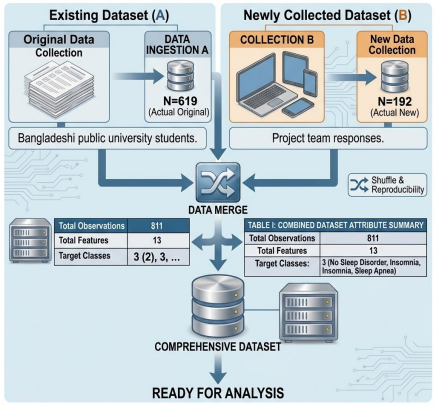


Fig. 1. Overview of the combined dataset

IV. EXPLORATORY DATA ANALYSIS

A. Dataset Overview

The merged dataset initially contains 811 observations and 13 columns, including 10 categorical variables and 3 numerical variables.

B. Missing-Value Analysis

Missing-value analysis revealed missing observations in: There were 60 missing Blood Pressure values, 74 missing Heart Rate values, and 38 missing target labels.

TABLE I
VARIABLE TYPES IN THE DATASET

Variable type	Variables
Categorical	Department, Gender, Sleep Duration, Quality of Sleep, Stress Level, BMI Category, Blood Pressure, Daily Steps, Academic Level, and Sleep Disorder.
Numerical	Age, Physical Activity Level, and Heart Rate.

C. Target Distribution

The data we're working with has three main categories: Insomnia, Sleep Apnea, and No Sleep Disorder. After removing 38 cases where the target was missing, we looked at 776 responses where the target was clear. Out of these, Insomnia was the most common problem, affecting 321 people, or about 41% of the group. Sleep Apnea was the next most common, affecting 249 people, or about 32% of the group. The least common was No Sleep Disorder, which applied to 203 people, or about 26% of the group.

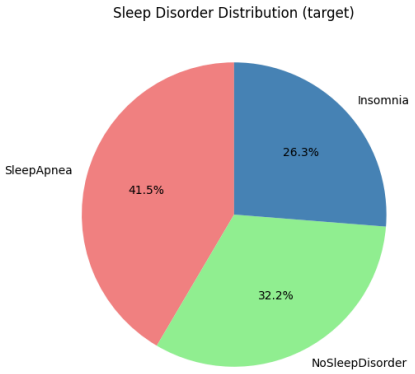


Fig. 2. Distribution of sleep-disorder target classes.

D. Demographic Analysis

The survey had a total of 811 people about 50%, said they were male. On the other hand, 376 people, or roughly 46%, said they were female. There were also 27 people, which is about 3%, who didn't want to say what their gender was. If we look at the different departments, we can see that the largest group is CSE, with 156 responses, making up 19.24% of the total. Following closely are EEE with 84 responses (10.36%), Civil Engineering with 63 (7.77%), ECE with 56 (6.91%), and Mechanical Engineering with 54 (6.66%).

The remaining departments (AIE, Economics, Mathematics, Statistics, Agriculture, Social Work, NE, FPE, DVM, and GDS) each contribute between 29 and 53 responses, together accounting for the rest of the sample.

E. Behavioral and Lifestyle Analysis

The dataset contains several behavioral and lifestyle variables, including Sleep Duration, Quality of Sleep, Stress Level, Physical Activity Level, and Daily Steps. Quality of Sleep and Stress Level are ordinal variables, meaning their

TABLE II
DEMOGRAPHIC DISTRIBUTION OF THE DATASET

Variable	Category	Count	Percentage
Gender	Male	408	50.31%
	Female	376	46.36%
	Prefer not to say	27	3.33%
Department	CSE	156	19.24%
	EEE	84	10.36%
	Civil Engineering	63	7.77%
	ECE	56	6.91%
	ME	54	6.66%

categories have a meaningful natural ordering; they are later converted into numerical scales, along with Academic Level and BMI Category. Sleep Duration and Daily Steps originally contain range-based responses. People do different amounts of physical activity, and we measured this in minutes. On average, they did about 62 minutes of activity. But the amount of time varied a lot - some people did as little as 10 minutes, while others did as much as 120 minutes.

Physical Activity Level is recorded as a continuous numerical variable (in minutes) rather than a categorical range, with a mean of 61.9 minutes and values ranging from 10 to 120 minutes.

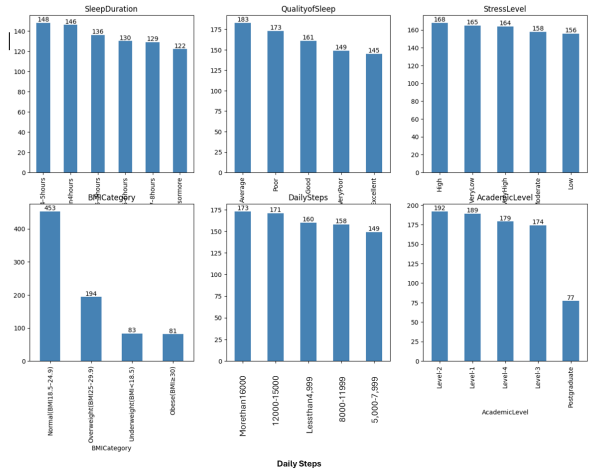


Fig. 3. Distribution of behavioral and lifestyle categorical variables by sleep-disorder class.

Table III presents the most frequent category for each selected behavioral and lifestyle variable.

TABLE III
MOST FREQUENT BEHAVIORAL AND LIFESTYLE CATEGORIES

Variable	Most Frequent Category	Count
Sleep Duration	4-5 hours	148
Quality of Sleep	Average	183
Stress Level	High	168
Daily Steps	More than 16,000 (gym session, power-walking, jogging/sports)	173

F. Clinical Analysis

The students' weight status is categorized based on their BMI, with the majority, 55.9%, having a normal weight, which accounts for 453 responses. After this, 23.9% of the students have an overweight body condition, representing 194 observations, whereas 10.2% have an underweight body condition, represented by 83 observations, and 10% of students are obese with 81 observations. In the case of blood pressure, initially, the data was combined for systolic and diastolic blood pressures into a string, having 60 missing observations. In order to enhance the usability of this data, it was divided into two different numeric variables for systolic and diastolic blood pressure. Heart rate data is treated as a continuous numerical feature. Initially, there were 74 missing responses for heart rate, which were handled by using median imputation. This imputation was calculated based solely on the training set, ensuring that the missing values were filled in a way that reflects the characteristics of the available data.

TABLE IV
CLINICAL ANALYSIS OF THE DATASET

Variable	Key observation or processing	Missing
BMI Category	Normal Weight is most frequent (453), followed by Overweight (194) Underweight (83), Obese (81)..	–
Blood Pressure	Combined systolic/diastolic values are separated into Systolic and Diastolic numerical features.	60
Heart Rate	Retained as a continuous numerical feature; missing values are handled using median imputation.	74

G. Correlation Analysis

A correlation matrix of the raw numerical variables (Age, Physical Activity Level, and Heart Rate) was examined ; these three variables show only weak pairwise correlations with each other (all below 0.09).

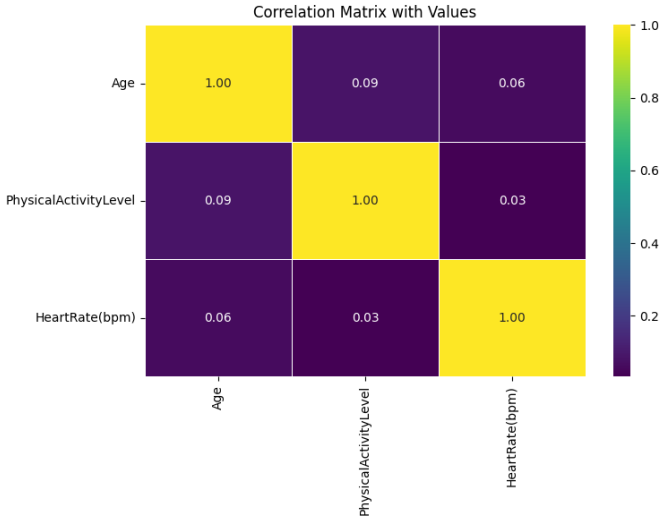


Fig. 4. Correlation matrix of all encoded features and the target.

V. PREPROCESSING

The preprocessing stage converts the raw survey responses into a numerical dataset suitable for machine learning.

A. Gender Encoding

The Gender variable is mapped to numerical codes (Male $\rightarrow$ 0, Female $\rightarrow$ 1, Prefer not to say $\rightarrow$ 2) rather than removing the "Prefer not to say" responses, so no rows are dropped at this step.

B. Duplicate Removal and Missing-Target Removal

Rows with a missing Sleep Disorder label (38 rows) are removed, since they cannot be used for supervised classification, leaving 773 observations.

C. Blood Pressure Transformation

The Blood Pressure variable is originally stored in the form Systolic/Diastolic (e.g., 120/80). The value is split into two separate numerical variables, Systolic and Diastolic, allowing the model to retain more information than a categorical bloodpressure representation.

D. Ordinal Encoding

Four naturally ordered variables are transformed into numerical scales: Quality of Sleep, Stress Level, Academic Level, and BMI Category.

E. Sleep Duration and Daily Steps Transformation

Sleep Duration and Daily Steps originally contain range-based responses. Numeric values are extracted from each range description and averaged to obtain the midpoint, producing the continuous features SleepDuration_hrs and DailySteps_num.

F. Department Encoding

Department is one-hot encoded with drop_first=True to remove a redundant reference category. All 15 departments are represented directly.

G. Train/Test Split, Imputation, SMOTE, and Scaling

The information is divided into two parts: one for training and one for testing, with 80% used for training and 20% for testing. This gives us 618 pieces of information for training and 155 for testing. The training information is used to fill in missing details about heart rate, systolic, and diastolic values. The missing values are filled with the middle values from the training information, which are 74.0 for heart rate, 114.0 for systolic, and 75.0 for diastolic. These filled-in values are then used for both the training and testing information. The goal is to understand sleep disorders, which are labeled as follows: Insomnia is labeled as 0, No Sleep Disorder as 1, and Sleep Apnea as 2. This labeling helps in analyzing the data to identify patterns and connections between different factors and sleep disorders. By doing this, we can better understand how different health metrics relate to sleep issues and use this knowledge to improve health outcomes. SMOTE is then applied to the training set only, balancing the class counts from Insomnia: 162, No Sleep Disorder:

199, Sleep Apnea: 257 to 257 observations per class (771 total). Features are standardized with StandardScaler for the Logistic Regression model; the tree-based models are trained on the unscaled, SMOTE-balanced features. After preprocessing, the final feature set contains 26 predictor columns (12 demographic/behavioral/clinical features plus 14 one-hot Department columns) and 1 target column.

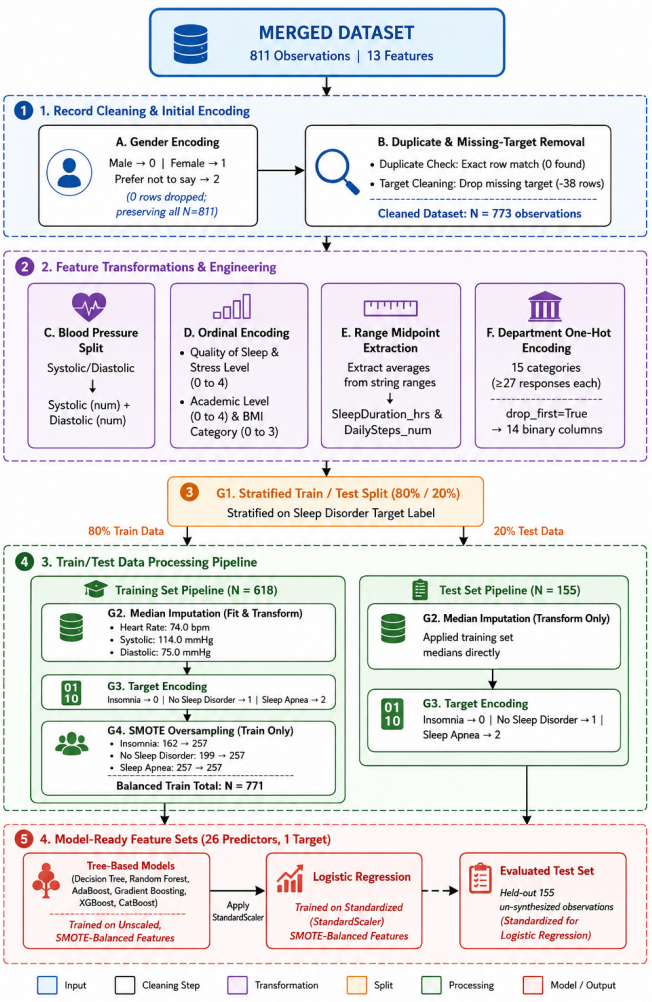


Fig. 5. Entire Preprocessing Pipeline

VI. MODELING

Seven classification algorithms were trained and evaluated: Decision Tree, Random Forest, AdaBoost, Gradient Boosting, Logistic Regression, XGBoost, and CatBoost. Decision Tree and Random Forest hyperparameters were tuned with GridSearchCV using 5-fold stratified cross-validation on the SMOTE-balanced training data. while Logistic Regression was trained on standardized features. All seven models were evaluated on the same held-out test set of 155 observations, which was separated before SMOTE and scaling.

A. Hyperparameter Tuning

The grid search was performed on the SMOTE-balanced training data. The remaining ensemble models used AdaBoost used 200 estimators with a learning rate of 0.5; Gradient Boosting used 200 estimators, a learning rate of 0.05, and a maximum depth of 3; XGBoost used 200 estimators, a learning rate of 0.05, a maximum depth of 5, and multi:logloss evaluation; and CatBoost used 200 iterations, a learning rate of 0.05, and a depth of 5.

TABLE V
GRID SEARCH RESULTS FOR DECISION TREE AND RANDOM FOREST

Model	Best Hyperparameters	Best CV Acc.
Decision Tree	criterion=entropy, min_samples_leaf=20	0.704
Random Forest	n_estimators=200, min_samples_leaf=1	0.750

Logistic Regression was trained with default parameters and max_iter=2000.

VII. RESULTS AND DISCUSSION

A. Overall Performance

CatBoost achieved the highest held-out accuracy (71.61%) and macro F1 (0.72), outperforming Gradient Boosting (69.68%) and Logistic Regression (69.03%). XGBoost produced the lowest test accuracy at 63.87%.

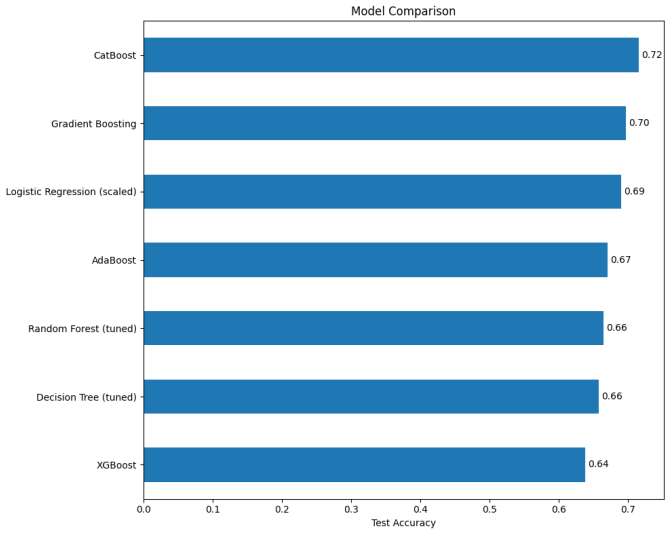


Fig. 6. Test accuracy comparison across the five models.

Although Random Forest achieved the highest cross-validation accuracy during grid search (0.750), its held-out test accuracy was 0.6645. In contrast, CatBoost reached 0.7161 on the same held-out test set, becoming the strongest overall model in the updated comparison.

B. Per-Class Performance and Confusion Matrices

Per-class results were evaluated for all seven models. CatBoost achieved 0.78 precision and 0.78 recall for Insomnia, 0.58 precision and 0.64 recall for No Sleep Disorder, and 0.80 precision and 0.73 recall for Sleep Apnea. No Sleep Disorder remains the most difficult class, while Sleep Apnea is generally identified more reliably.

AdaBoost exhibits a distinctive error pattern, favoring precision over recall for Insomnia and Sleep Apnea (0.86 and 0.82

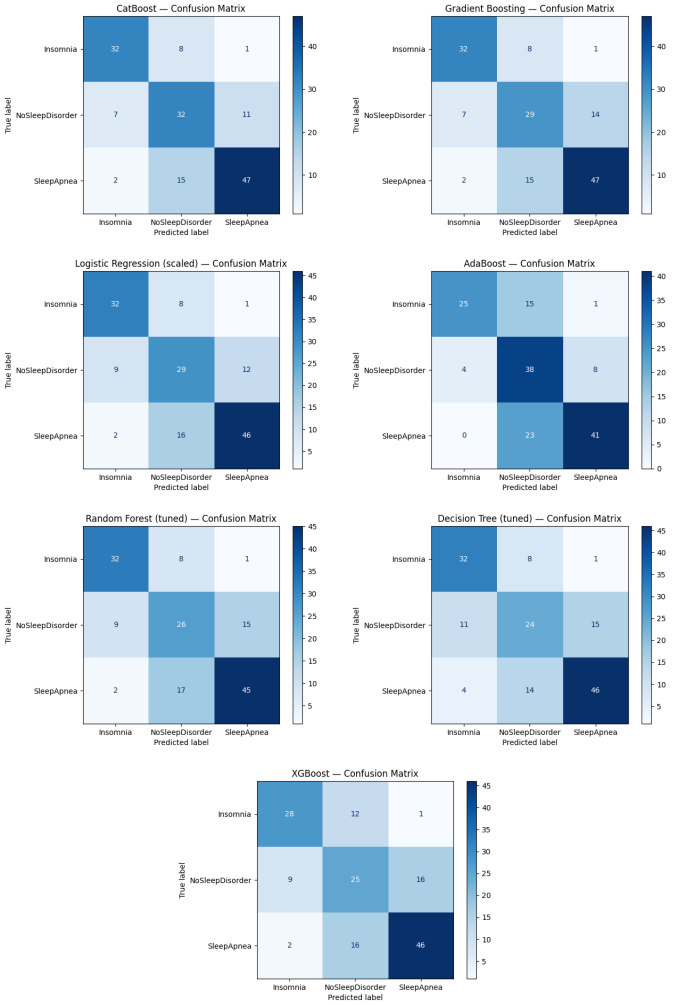


Fig. 7. Confusion matrices on the test set: (a) CatBoost, (b) Gradient Boosting, (c) Logistic Regression, (d) AdaBoost, (e) Random Forest, (f) Decision Tree, and (g) XGBoost.

precision respectively) at the cost of substantially lower recall for those classes, while overpredicting the No Sleep Disorder class (0.76 recall).

C. Discussion

Sleep problems can be predicted by looking at how long someone sleeps and how stressed they feel. For university students in Bangladesh, asking about sleep and stress could be a simple way to screen for sleep issues. The current approach is not ideal, however, and could be improved. It's hard to tell the difference between people who don't have sleep problems and those who have sleep apnea. But we may need more information or to research a larger group of non-sleeping patients to improve it. This might enable us to make more with less. This could help us make more accurate predictions and improve the overall accuracy of the test.

VIII. CONCLUSION

This project was an on-going project of the Bangladeshi university universities. A sleep-disorder dataset of new responses was collected, created a combined data set with 811 records, and created a complete machine learning pipeline by working on exploratory data analysis. Class imbalance and SMOTE preprocessing were done for analysis of survey. The study incorporates balancing and comparative evaluation. Seven classification algorithms were evaluated such as Decision Tree, Random Forest, Decision Trees, Random Forests, Gradient Boosting, XGBoost, AdaBoost, Logistic Regression, and CatBoost. CatBoost performed the best where the accuracy of the test is 71.6% and the macro f1 score is 0.72. The second highest was Gradient Boosting with 69.7%. Future work could investigate external validation, additional clinical and Different kinds of behavioral variables, model calibration and explainable-AI.

REFERENCES

[1] R. Alazaidah, G. Samara, M. Aljaidi, M. Haj Qasem, A. Alsarhan, and M. Alshammari, "Potential of machine learning for predicting sleep disorders: A comprehensive analysis of regression and classification models," *Diagnostics*, vol. 14, no. 1, Art. no. 27, 2024, doi: 10.3390/diagnostics14010027.

[2] L. Ravikumar and P. K. Bhansali, "Sleep disorder prediction using machine learning," *International Journal of Engineering Technology Research & Management*, vol. 9, no. 7, 2025, doi: 10.5281/zenodo.16569127.

[3] M. Kumar, B. Ahmed, H. M. Mishra, A. K. Jha, P. K. Sikarwal, and S. Rampal, "Predictive sleep disorder modelling: Using machine learning with lifestyle and sleep health data," in *Proc. 2024 Int. Conf. Advances in Computing, Communication and Applied Informatics (ACCAI)*, 2024, doi: 10.1109/ACCAI61061.2024.10602153.

[4] S. Goodyear and A. Alsammani, "Predicting sleep disorders using machine learning: A comparative analysis," in *Computational Science and Its Applications – ICCSA 2025*, Lecture Notes in Computer Science, pp. 106–121, 2025, doi: 10.1007/978-3-031-97000-9_7.

[5] T. O. Adeyemi, "Optimized machine learning algorithms for the classification and diagnosis of sleep disorders," medRxiv, 2025, doi: 10.1101/2025.10.07.25337263.

[6] T. S. Alshammari, "Applying machine learning algorithms for the classification of sleep disorders," *IEEE Access*, vol. 12, 2024. [Online]. Available: https://jpinfotech.org/sleep-disorder-prediction-using-machine-learning/

[7] Y. S. Taspinar and I. Cinar, "Prediction of sleep health status, visualization and analysis of data," in *Proc. Int. Conf. Advanced Technologies (ICAT)*, 2023, doi: 10.58190/icat.2023.13.

[8] A. Taher and W. I. Z. Ayon, "Exploring sleep disorders: A comparative analysis of machine learning algorithms on sleep health and lifestyle data," in *Proc. 2024 IEEE Int. Conf. Power, Electrical, Electronics and Industrial Applications (PEEIACON)*, 2024, doi: 10.1109/PEEIACON63629.2024.10800593.

[9] A. A. Alghwiri, F. Almomani, A. A. Alghwiri, and S. L. Whitney, "Predictors of sleep quality among university students: The use of advanced machine learning techniques," *Sleep and Breathing*, vol. 25, pp. 1119–1126, 2021, doi: 10.1007/s11325-020-02150-w.

[10] N. A. Lopa, S. M. Opu, S. Roy, M. I. Z. Soma, and P. Bhowmik, "A Cross-Sectional Dataset of Sleep Disorder Indicators Among Bangladeshi University Students," Mendeley Data, V1, 2026, doi: 10.17632/9djgjvtj8d.1.